

A data project about Boston's gas pipes

Fixing Boston's old gas pipes

Repair, replace, or switch to electric heat? A street-by-street look.

Cathy Kam

The problem

Much of Boston's gas pipe is old cast iron and steel that can leak

34%

of Boston Gas's main pipes are made of leak-prone materials, such as cast iron and unprotected steel.

9,077

gas leaks were still open across Massachusetts at the end of 2024.

Why it matters

Natural gas is mostly methane, a powerful greenhouse gas. Leaks waste gas and can be a safety risk.

What's been done

Since 2015, Massachusetts has paid to replace leaky pipes

\$4.75B

spent from 2015 to 2024 under the Gas System Enhancement Program (GSEP).

2,411

miles of leak-prone gas main replaced across the state in that time.

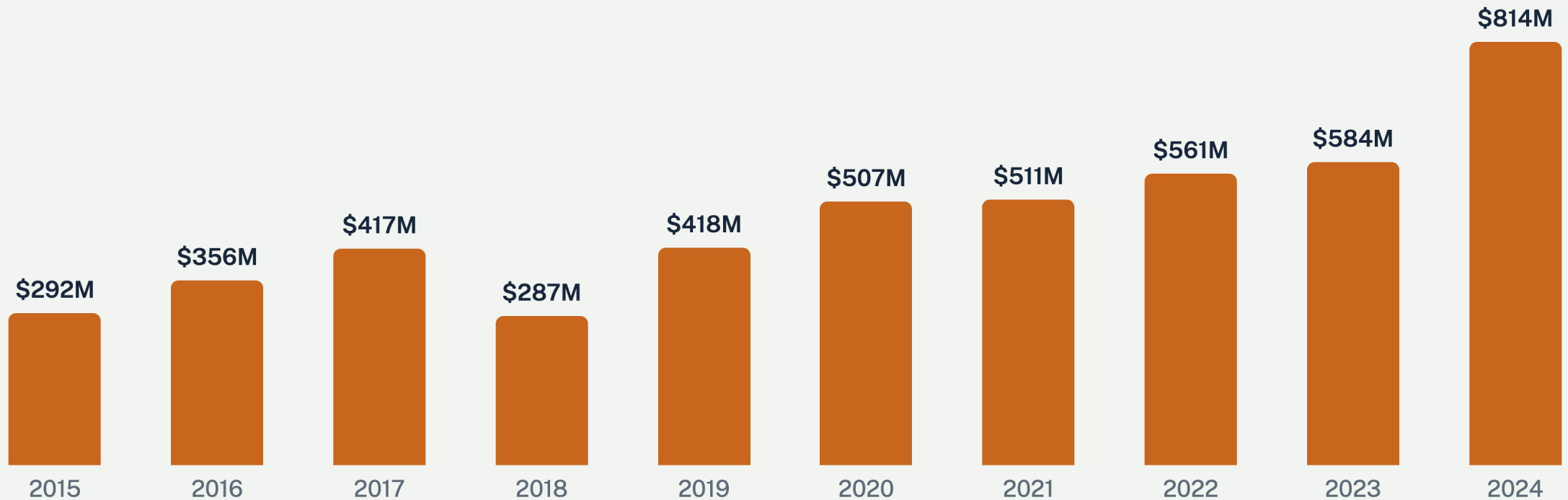
Who pays

The cost is added to customers' gas bills over many years. It is not paid from taxes.

The cost

Yearly spending nearly tripled

Money spent each year on replacing leak-prone pipe, statewide

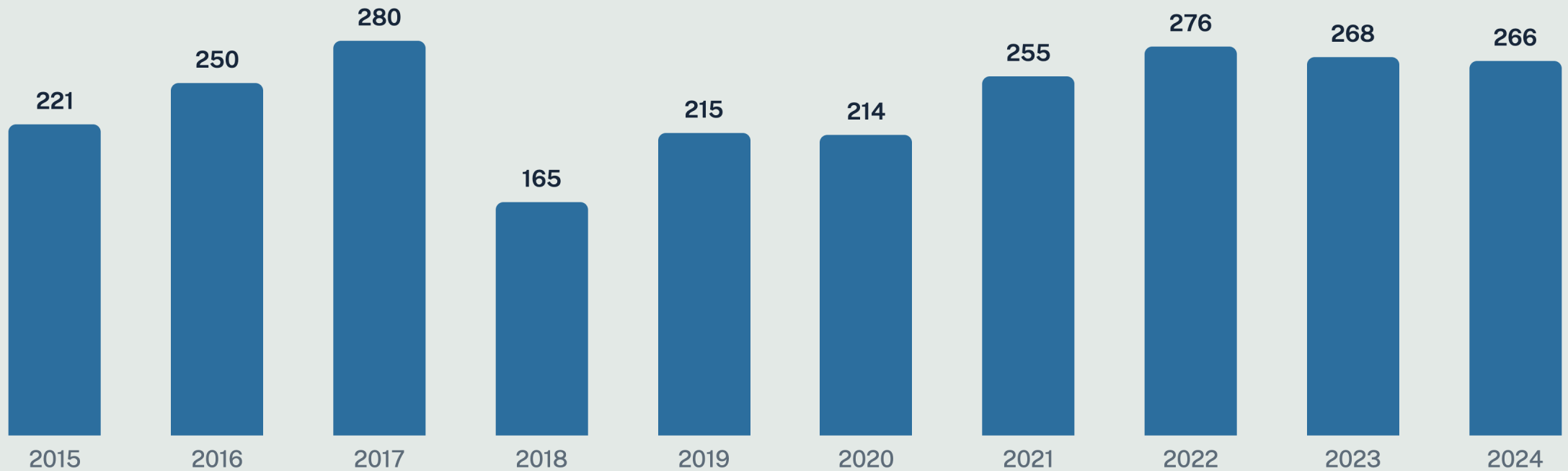


Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025). Includes mains, service lines and related work; not adjusted for inflation.

The progress

But the miles replaced each year barely changed

Miles of leak-prone gas main replaced each year, statewide



Source: Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

Cost per mile

\$3.06 million

is what replacing one mile of gas pipe cost in 2024, up from \$1.32 million in 2015.

Calculated from DPU-reported spending and miles. Massachusetts DPU, Report to the Legislature on Natural Gas Leaks, D.P.U. 25-GLR-01 (Dec. 31, 2025)

Did it help?

Yes: open gas leaks fell by more than half

End of 2014

20,775

open leaks statewide



End of 2024

9,077

open leaks statewide, 56% fewer

The new question

In 2025, regulators told gas companies to look at cheaper options

One alternative

Instead of laying new gas pipe under a street, help the homes on that street switch to electric heat pumps, then retire the old pipe.

A real test

Gas companies reviewed 500 possible alternative projects for 2026. None were judged worth pursuing.

So which is cheaper, street by street?

How I answered it

Four steps, all with public data

1

Gather the data

About 17,800 Boston street segments, joined with building records, energy use, Census data and state reports.

2

Look for patterns

Where streets are oldest, most dependent on gas, and home to the most people.

3

Build the models

Group similar streets, flag likely old pipe, and compare the cost of each option.

4

Share the results

An interactive map where anyone can explore streets and change the assumptions.

Finding 1

The tipping point is about 8.6 homes per 100 meters of street

New pipe costs about **\$3.06 million** per mile.

A heat pump for one home costs about **\$22,000** (an estimate; prices vary).

\$3.06 million buys heat pumps for about **139 homes**, which works out to 8.6 homes for every 100 meters of street.

Fewer homes

Switching the homes to heat pumps costs less than new pipe.

More homes

Laying new pipe costs less, because its cost is shared by more homes.

Finding 2

On most of Boston's oldest streets, new pipe is the lower-cost option

351 miles of Boston streets likely sit on old, leak-prone pipe



35 miles: cheaper to electrify

316 miles: cheaper to replace the pipe

2,800

homes, roughly, live on the streets where electrifying wins.

19

homes per 100 meters is typical on Boston's dense triple-decker and rental streets.

4%

overall saving from electrifying the small share of qualifying streets.

Finding 3

Most streets where electrifying wins are in environmental-justice neighborhoods

7
8
%

of those streets are in neighborhoods the state identifies as environmental-justice communities, based on income, race and English-language ability.

What that means

Any program there would need income-based financial help, and protections for renters, who don't control their building's heating system.

Not evaluated in this analysis

Shared systems such as networked geothermal remain unproven at scale

What it is

One underground loop of water pipes serves many buildings, drawing on the ground's steady temperature for heating and cooling.

What pilots show so far

Pilots are few and early. National Grid canceled a planned pilot in Lowell over high cost per customer; its Dorchester project serves 129 public housing units.

Still waiting for pilot cost data; this analysis will evaluate shared systems once it is available.

Honest limits

What this analysis can't tell us yet

Exact pipe locations

Old pipe is estimated from building age, not from utility pipe or leak records.

Local costs

It uses the statewide cost per mile. Boston's busy streets may cost more.

The whole network

Removing gas from one street can affect its neighbors, which the model can't see.

Heat pump prices

At \$15,000 per home, savings rise to about 7%; at \$40,000, they fall to about 2%.

Takeaways

- Replacing pipe has cut leaks, but the cost per mile has more than doubled.
- Switching homes to heat pumps is cheaper only on streets with few homes.
- On most old streets, targeted replacement prioritized by risk remains the cost-effective path.
- Networked geothermal: still waiting for pilot data before it can be evaluated.

Explore the interactive app:

boston-gas-triage-kljspjcnsvm6y8pc8uv3sp.streamlit.app

Cathy Kam. Built with public data; not an official DPU or utility analysis.